

Project Analysis Report: Hormonal Study

1. Descriptive Statistics

The following table summarizes the central tendency and dispersion for all hormones across groups 0 and 1.

Hormone	Group 0 (Mean/SD)	Group 1 (Mean/SD)	Shapiro (G0)	Levene p	Wilcox p	Final Choice
hormone1	M=1.82, SD=1.58	M=1.66, SD=0.86	6.23e-44	0.087	0.7587	Wilcoxon
hormone2	M=375.15, SD=255.41	M=411.96, SD=270.89	8.29e-37	0.979	0.0054	Wilcoxon
hormone3	M=8.99, SD=11.58	M=9.91, SD=11.79	6.06e-44	0.577	0.1346	Wilcoxon
hormone4	M=7.51, SD=10.92	M=9.40, SD=13.44	7.42e-50	0.063	0.1482	Wilcoxon
hormone5	M=335.61, SD=332.26	M=276.45, SD=235.44	4.15e-44	0.165	0.0037	Wilcoxon
hormone6	M=78.89, SD=53.99	M=75.75, SD=48.68	1.77e-28	0.534	0.8160	Wilcoxon
hormone7	M=5.63, SD=3.90	M=5.65, SD=3.97	1.21e-31	0.799	0.9206	Wilcoxon
hormone8	M=180.02, SD=87.23	M=153.76, SD=70.96	6.21e-21	0.071	0.0004	Wilcoxon
hormone10_generated	M=0.76, SD=8.30	M=0.51, SD=0.37	4.74e-57	0.691	0.7905	Wilcoxon

2. Assumption Testing

Normality: Shapiro-Wilk tests for all hormones resulted in $p < 0.05$, indicating non-normal distributions. Non-parametric methods are mandatory.

Variance: Levene's test showed $p > 0.05$ for most variables, suggesting homogeneity of variance across groups.

3. Comparative Analysis (Brunner-Munzel vs. Others)

We calculated p-values using t-tests, Wilcoxon, and Brunner-Munzel. Given the violation of normality, the **Wilcoxon Rank-Sum** or **Brunner-Munzel** tests are the valid choices. Specifically, Brunner-Munzel is preferred for its robustness against unequal variances, though Wilcoxon is highly reliable here due to equal variances.

4. Correlation Analysis

Spearman correlation was employed. Group 0 shows strong positive co-regulation among hormones 3, 4, and 5. Group 1 maintains some of these links but shows higher variance in correlation strength.